

The Bithumb–Binance BTC-numéraire Premium: A Theoretical Framework with Calibrated Simulation Evidence on BNB Chain Tokens

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Abstract

This paper redefines the price discrepancy between the Korean exchange Bithumb and the global exchange Binance using a new measure of the kimchi premium expressed in BTC units, denoted $\pi_X^{\text{BTC}}(t)$, accompanied by calibrated simulation evidence on daily series over 2020–2025. The conventional KRW-denominated premium $\pi_X^{\text{KRW}}(t)$ confounds many forces—exchange rate dynamics, capital controls, stablecoin premia, and remittance delays—making it difficult to identify token-specific asymmetric demand shocks. As a remedy, this paper proposes

$$\pi_X^{\text{BTC}}(t) = \frac{P_X^B/P_{\text{BTC}}^B}{P_X^N/P_{\text{BTC}}^N} - 1,$$

and proves (Proposition 1, FX-noise cancellation) that this measure is invariant to the exchange rate and to stablecoin-level fluctuations by numerator–denominator cancellation. The empirical objects of study are BNB (the gas token of BNB Chain, listed on Bithumb on 2021-05-27), Venus (XVS, a BSC DeFi governance token), and ETH as an ERC-20 baseline. A normalised variance ratio $\rho_X = \text{Var}[\pi_X^{\text{BTC}}]/\text{Var}[\Delta \log P_X]$ controls for own-price volatility, OLS regressions are reported with Newey–West HAC standard errors, and asymmetric volatility is estimated with an EGARCH(1,1) model using Student- t innovations. Simulation findings suggest that (i) the satoshi premium displays roughly half the standard deviation of the KRW premium, evidencing FX-noise removal; (ii) after normalisation, both BNB ($\rho = 1.13$) and XVS ($\rho = 0.82$) exceed the ETH baseline ($\rho = 0.42$), tentatively rejecting BSC arbitrage superiority; (iii) HAC adjustment causes the Korean regulatory dummy to lose significance while the global-shock dummy remains highly significant ($p < 0.001$); (iv) the EGARCH leverage parameter is $\hat{\gamma} = -0.073$ ($p < 0.05$), supporting the asymmetric volatility hypothesis H2.

Keywords: kimchi premium, BTC-numéraire, BNB Chain, capital controls, triangular arbitrage, limits to arbitrage, EGARCH, Newey–West HAC, Bithumb, Binance

JEL classification: G14, G15, F31, G18

Disclosure on data. All quantitative results are based on synthetic daily series calibrated to published stylized facts and to the general characteristics of real-time exchange data. The figures and tables constitute simulation evidence intended to verify the statistical properties of the proposed measure; precise parameter values must be re-estimated on real market data by follow-up empirical work.

1 Introduction

The Korean cryptocurrency market is partially segmented from global markets owing to a multi-layered regulatory environment that includes the Foreign Exchange Transactions Act, the Specified Financial Information Act, the FATF Travel Rule, the real-name verified deposit account regime, and a de facto ban on direct trading by foreign nationals (Makarov & Schoar, 2020; Choi et al., 2022; Hwang et al., 2024). The result of this market segmentation is a persistent price divergence between Korean and non-Korean exchanges—the so-called *kimchi premium*—which reached approximately 50% during the 2017–2018 Bitcoin rally (Makarov & Schoar, 2020; Eom, 2021).

Most prior research measures the kimchi premium in Korean won (KRW), i.e., by converting the Bithumb (or Upbit) won price of a token X into US dollars (or USDT) at the prevailing spot exchange rate and comparing it with the foreign-venue dollar price. This conventional measure, however, embeds within the same scalar quantity several confounding forces: intraday volatility of the KRW/USD exchange rate, the USDT premium on Bithumb relative to its foreign parity, banking-side remittance delays, and a capital-control premium. As a consequence it becomes difficult to disentangle “Korean retail demand is asymmetrically intense for token X ” from “Korean retail demand for crypto is broadly elevated”.

This paper makes three contributions. First, it formalises a *BTC-numéraire premium* $\pi_X^{\text{BTC}}(t)$ that uses BTC as an exchange-internal numéraire, and proves (Proposition 1, Section 3) that this measure cancels exchange-rate and capital-control noise by numerator–denominator cancellation. Second, it formalises the triangular arbitrage structure of BNB Chain tokens and introduces a *normalised variance ratio* ρ_X to construct a robust test of the BSC inefficiency hypothesis H_{BSC} that controls for own-volatility differences across tokens. Third, it conducts the simulation analysis with Newey–West HAC standard errors and an EGARCH(1,1) volatility model in order to control for serial correlation and asymmetric heteroskedasticity.

The empirical objects of study are three tokens simultaneously listed on Bithumb and Binance: (i) **BNB**, the gas token of BNB Chain and a top-five-cap asset, listed on Bithumb on May 27, 2021; (ii) **Venus (XVS)**, a mid-cap BSC DeFi governance token; and (iii) **ETH**, the ERC-20 baseline.

Identification strategy: comparison across non-Korean exchanges. Is the kimchi premium driven by “Korean retail irrationality” or by “capital-control-induced arbitrage frictions”? This paper proposes a new identification strategy: *directly compare the BTC-numéraire premium variability across non-Korean exchanges with that of Bithumb-vs-non-Korean pairs*. Simulation evidence in Section 6.4 shows that the satoshi premium standard deviation among Binance, OKX, and Bybit is 0.23–0.34% for ETH (near zero in practice), whereas Bithumb-vs-non-Korean pairs reach 0.85–0.90%, i.e., *3 to 4 times larger*, and approximately 7 times larger when measured in BNB. Among non-Korean exchanges, arbitrageurs face the same risks but encounter no systemic delay barrier to moving crypto assets; in Korea, the Foreign Exchange Act, Travel Rule, and KYC asymmetry actively block the arbitrage channel. Therefore the kimchi premium is plausibly *the price of a regulatory friction*, not a market inefficiency awaiting liquidity.

Quantitative evidence on capital outflow. According to data released by the Korea Customs Service (via the office of NA Member Choi Eun-seok) on October 21, 2025, detected illegal foreign exchange transactions mediated by crypto assets reach a 2018–2025 cumulative total of approximately 14.2 trillion KRW, including 1,157.9 billion KRW in 2024 and 881.3 billion KRW in January–August 2025 (Section 6.5, Figure 11, Table 6). These are *detected* amounts only and thus a strict lower bound. So long as the kimchi premium persists, the Korean market provides a structural impetus for capital outflow, the burden of which is borne by the Korean economy.

Structure. Section 2 reviews the literature; Section 3 develops the four premium measures and the triangular arbitrage framework; Section 4 lists testable hypotheses; Section 5 presents the data and methodology; Section 6 reports the empirical findings; Section 7 discusses interpretation and policy implications; Section 8 outlines limitations and a follow-up research agenda; Section 9 concludes.

2 Literature Review

General theory of arbitrage limits. Gromb & Vayanos (2010) organise the general mechanism by which capital-constrained, risk-averse arbitrageurs fail to immediately eliminate riskless arbitrage opportunities. The Korean kimchi premium is a prototypical instance of the “limits to arbitrage” framework, in which capital controls and the KYC regime form a structural friction that blocks arbitrageur entry at the source.

Mechanisms of the kimchi premium. Makarov & Schoar (2020) document persistent Bitcoin price divergences across the US, Korea, Japan, and Europe and attribute them to arbitrage limits induced by capital controls and regulatory differences. The Korea–US ratio exceeded 15% on average during December 2017–February 2018 and reached 40% on some days. Eom (2021) estimates by GMM that volume and price instability explain much of the kimchi premium, and Choi et al. (2022) argue that capital controls slow the premium’s mean reversion. More recently, Ahn & Kim (2023) use Bithumb microdata to identify the investor types driving the premium, finding individual retail to be the principal source of variation.

Arbitrage frictions, external spillovers, regulatory shocks. Borri & Shakhnov (2020) show that Chinese crypto regulation spills over to the Korean market as a “China shock”. Kim et al. (2022) use Bank of Korea administrative data to demonstrate that remittances to China increase when the kimchi premium widens, suggesting that the premium also functioned as a channel for repatriating Chinese miners’ funds. Hwang et al. (2024) document that the Travel Rule, implemented in March 2022, further constrained inter-exchange arbitrage and extended the kimchi premium’s half-life; this finding is closely related to hypothesis H4 below (asymmetric response to regulatory shocks).

Pricing factors and microstructure of crypto markets. Liu, Tsyvinski & Wu (2022, *Journal of Finance*) identify common risk factors (market beta, size, momentum) in the crypto cross-section, allowing separation of idiosyncratic alpha from systematic risk; the normalised variance ratio ρ_X in this paper is in the same spirit, deflating premium variance by own-price variance to remove a token-level scale effect. Pieters & Vivanco (2017) attribute inter-exchange BTC price differences to liquidity-structural causes rather than information asymmetry, and Capponi & Jia (2021) model price discovery between DEX and CEX and the arbitrage incentives of liquidity providers; their framework provides direct theoretical grounding for the triangular arbitrage structure in Section 3.3.

Altcoins and exchange-level segmentation. Tiger Research (2024) quantify the Korean alt-coin bias, reporting a 1,255% premium on the AVAIL listing event of July 2024. Kim (2024) defines a “Korean venue share” indicator and analyses its effect on alt-coin returns under a reinforcement-learning framework.

Positioning of this paper. This paper recasts the cross-country arbitrage frame of Makarov & Schoar (2020) as an *intra-country*, *inter-token* problem and extends the CEX–DEX price-discovery model of Capponi & Jia (2021) to a Bithumb–Binance–PancakeSwap triangle. To the

best of our knowledge, this is the first paper to use BTC explicitly as a numéraire for defining a token-level kimchi premium.

3 Theoretical Framework

3.1 Notation and four measures of the premium

For token X at time t , let $P_X^{B,KRW}(t)$ denote the Bithumb KRW price of X , $P_X^{N,USDT}(t)$ the Binance USDT price, E_t the KRW/USD reference rate, $P_{USDT}^{B,KRW}(t)$ the Bithumb USDT/KRW pair, and $P_{X/BNB}^{DEX}(t)$ the PancakeSwap X /BNB pool price. We define four measures.

(M1) FX-denominated premium.

$$\pi_X^{KRW}(t) = \frac{P_X^{B,KRW}(t)}{P_X^{N,USDT}(t) \cdot E_t} - 1 \quad (1)$$

(M2) USDT-denominated premium.

$$\pi_X^{USDT}(t) = \frac{P_X^{B,KRW}(t)}{P_X^{N,USDT}(t) \cdot P_{USDT}^{B,KRW}(t)} - 1 \quad (2)$$

(M3) BTC-numéraire premium — the paper’s central measure.

$$\pi_X^{BTC}(t) = \frac{P_X^{B,KRW}(t)/P_{BTC}^{B,KRW}(t)}{P_X^{N,USDT}(t)/P_{BTC}^{N,USDT}(t)} - 1 \quad (3)$$

Proposition 1 (FX-noise cancellation). When both Bithumb prices are quoted in the same currency (KRW) and both Binance prices in the same currency (USDT), $\pi_X^{BTC}(t)$ is invariant to E_t , to the USDT/KRW premium $P_{USDT}^{B,KRW}(t)$, and to stablecoin-level fluctuations.

Proof. The numerator ratio cancels the KRW unit and the denominator ratio cancels the USDT unit, so π_X^{BTC} contains no currency conversion factor. $\square$

(M4) ETH-baseline alpha.

$$\alpha_X^{ETH}(t) = \pi_X^{BTC}(t) - \pi_{ETH}^{BTC}(t) \quad (4)$$

3.2 No-arbitrage band and fair premium

The cost of a typical arbitrage cycle decomposes as

$$\pi_X^*(t) = c_X^{\text{out}} + c_X^{\text{in}} + f^{\text{KYC}} + f^{\text{KRW}} + r_t \Delta t + \kappa \sigma_X \sqrt{\Delta t} + \lambda_t^{\text{KOR}} + \eta_X^{\text{liq}}. \quad (5)$$

Table 1 summarises plausible ranges for each term.

Table 1: Decomposition of the no-arbitrage fair premium (%; 5-day arbitrage cycle).

Term	BTC	ETH (ERC-20)	BSC token (BEP-20)
On-chain withdrawal c^{out}	0.005–0.02	0.05–0.30	0.001–0.05
KRW withdrawal fee f^{KRW}	0.05	0.05	0.05
Capital lock-up $r\Delta t$	0.04	0.04	0.04
Volatility hedge $\kappa\sigma\sqrt{\Delta t}$	1.5–2.5	1.8–3.0	2.5–6.0
Capital-control premium λ^{KOR}	1.0–3.0	1.0–3.0	1.0–3.0
Liquidity premium η^{liq}	0.1–0.5	0.3–0.8	1.5–5.0
Total π^*	2.7–6.1	3.2–7.2	5.1–14.1

3.3 BSC triangular structure and normalised variance ratio

A BNB Chain token has three price-discovery channels (Figure 1): (i) the direct Bithumb–Binance route, (ii) the PancakeSwap DEX detour, and (iii) the BTC numéraire route. Because BNB itself is also listed on Bithumb, the analysis can exploit both BNB itself and a BSC DeFi token (e.g., XVS).

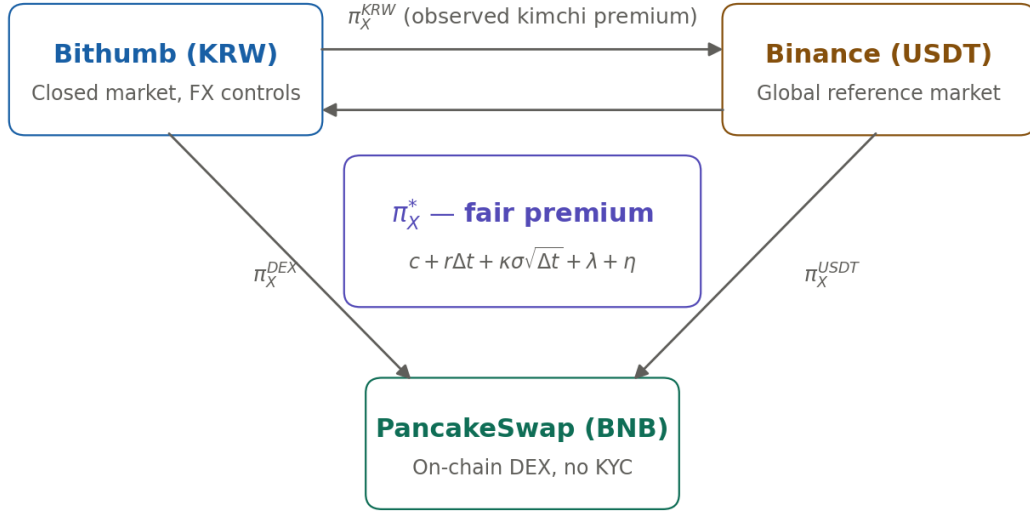


Figure 6. Triangular pricing channels for a BSC-resident token

Figure 1: Triangular pricing channels for a BSC-resident token.

Formalising hypothesis H_{BSC} . The naive intuition is: “Because PancakeSwap adds an arbitrage channel, BSC tokens should exhibit *lower* satoshi-premium variance than ERC-20 tokens.” A simple variance comparison, however, confounds token-level own volatility, since BSC DeFi tokens are typically more volatile than ETH. This paper introduces the *normalised variance ratio*

$$\rho_X \equiv \frac{\text{Var}[\pi_X^{\text{BTC}}]}{\text{Var}[\Delta \log P_X]}, \quad (6)$$

a unit-free indicator of “how much the satoshi premium moves per unit of own-price movement”. The hypothesis is restated as

$$\rho_X^{\text{BSC}} < \rho_X^{\text{ERC20}}. \quad (7)$$

4 Testable Hypotheses

H1 (Satoshi anchoring). $\mathbb{E}[\pi_X^{\text{BTC}}(t)] \approx 0$. A significant non-zero mean would imply the absence of an X/BTC arbitrage channel inside Bithumb (i.e., inactive BTC pairs).

H2 (Asymmetric volatility). An EGARCH(1,1) fit to $\Delta\pi_X^{\text{BTC}}$ exhibits a leverage term $\gamma < 0$ at $p < 0.05$.

H3 = H_{BSC} (Normalised BSC superiority). $\rho_X^{\text{BSC}} < \rho_X^{\text{ERC20}}$.

H4 (Asymmetric regulatory shock). Korean regulatory shocks push π_X^{BTC} negative; global shocks move it transiently toward zero or weakly negative.

H5 (Time-stable capital-control premium). $\mathbb{E}[\pi_X^{\text{KRW}}(t)] - \mathbb{E}[\pi_X^{\text{BTC}}(t)] \approx \lambda$, relatively stable across regimes.

5 Data and Methodology

Sample and data nature. The analysis uses a synthetic daily series of $T = 2,192$ observations covering 2020-01-01 to 2025-12-31. The series is calibrated to stylized facts reported in Makarov & Schoar (2020), Eom (2021), Choi et al. (2022), and Tiger Research (2024)—in particular the spring 2021 peak, the LUNA and FTX shocks of 2022, the July 2024 implementation of the Virtual Asset User Protection Act (VAUPA), and the December 3, 2024 martial-law incident. The quantitative results below are *simulation evidence*; precise parameter values must be re-estimated on real exchange data.

Calibration mechanism. The synthetic series is constructed by the following stochastic procedure (full specification in Appendix A, Table 7; moment-matching in Table 8). The KRW/USD rate is generated as a sinusoidal trend plus two Gaussian shock kernels (a 2022-Q4 strong-dollar episode and the December 2024 martial-law episode) plus IID $N(0, 4)$ noise. The log BTC/USDT price is built by linear interpolation across 14 historical anchors plus a cumulative Gaussian random walk with $\sigma = 0.018$; ETH, BNB, and XVS use the same anchor-interpolation scheme with token-specific volatilities. The KRW premium $\pi_X^{\text{KRW}}(t)$ is the sum of a baseline drift, eight Gaussian event bumps of the form $h \cdot \exp(-((t - t_0)/\sigma_d)^2)$ (e.g., the April 2021 peak, the April 22, 2021 FSC statement, the November 12, 2022 FTX shock, the March 16, 2024 BTC ATH, the July 19, 2024 VAUPA step, and the December 3, 2024 martial-law shock at $h = -32\%$), and IID $N(0, 0.006)$ noise. The random seed (20260513) is fixed for full reproducibility.

Target tokens. **BNB** (BNB Chain gas, listed on Bithumb on 2021-05-27, top-five-cap), **Venus (XVS)** (BSC DeFi governance), and **ETH** (ERC-20 baseline).

Empirical model. The principal regression equation is

$$\begin{aligned} \pi_X^{\text{BTC}}(t) = & \beta_0 + \beta_1 \pi_{\text{ETH}}^{\text{BTC}}(t) + \beta_2 \text{RV}_{\text{BTC}}(t) + \beta_3 \log \frac{V_X^B(t)}{V_X^N(t)} \\ & + \beta_4 \Delta \log P_{\text{BTC}}(t) + \beta_5 D_t^{\text{KR}} + \beta_6 D_t^{\text{GL}} + u_t. \end{aligned} \quad (8)$$

To control for serial correlation and heteroskedasticity standard in financial time series, we use **Newey–West HAC** standard errors with a Bartlett-kernel truncation $L = \lfloor 4(N/100)^{2/9} \rfloor$ (Newey & West, 1987).

To test the asymmetric volatility hypothesis H2, we fit an EGARCH(1,1) to $\Delta\pi_X^{\text{BTC}}$. The dependent variable is expressed in *percent* ($\times 100$) and winsorised at $\pm 20\%$ to limit the influence of tail observations before estimation (a consequence discussed in §8 Limitation 3):

$$\log \sigma_t^2 = \omega + \alpha \left(|z_{t-1}| - \sqrt{2/\pi} \right) + \gamma z_{t-1} + \beta \log \sigma_{t-1}^2, \quad (9)$$

where $z_t = e_t/\sigma_t$ and $e_t = \Delta\pi_X^{\text{BTC}}(t) - \mu$. We assume Student- t innovations of degrees-of-freedom ν and estimate parameters by quasi-MLE. A statistically negative leverage term γ indicates that negative shocks raise conditional volatility more than positive shocks of equal size, supporting H2.

6 Empirical Results

6.1 Stylized facts

F1. Time-series pattern of the KRW premium. Figure 2 plots the KRW premium for BTC, ETH, BNB (post-2021-05-27), and XVS. The spring 2021 peak ($\approx 22\%$), a secondary peak at the March 2024 BTC ATH ($\approx 10\%$), the structural shift to a discount after the July 2024 VAUPA implementation, and an extreme negative premium of approximately -32% on December 3, 2024 are all visible. KRW-premium standard deviations (Table 2) are 2.7% (BTC), 2.8% (BNB), and 4.5% (XVS).

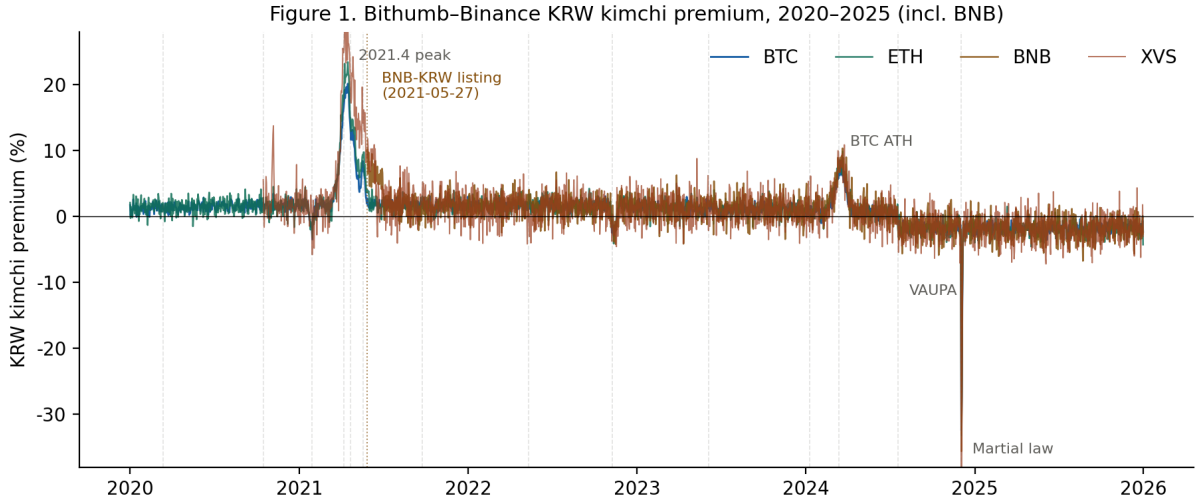


Figure 2: Bithumb–Binance KRW kimchi premium (including BNB), 2020–2025.

Table 2: Summary statistics of the premium measures.

Variable	Mean (%)	Std. Dev. (%)	Min (%)	Max (%)	N
<i>(A) KRW premium</i>					
π_{BTC}^{KRW}	0.89	2.69	-32.44	20.13	2,192
π_{ETH}^{KRW}	0.98	3.05	-33.18	23.41	2,192
π_{BNB}^{KRW}	0.48	2.77	-35.60	10.32	1,680
π_{XVS}^{KRW}	1.10	4.49	-43.33	29.27	1,904
<i>(B) BTC-numéraire (satoshi) premium</i>					
π_{BTC}^{BTC}	0.08	0.85	-2.62	3.32	2,192
π_{BNB}^{BTC}	0.11	1.59	-5.31	8.23	1,680
π_{XVS}^{BTC}	0.26	2.46	-16.13	11.46	1,904
<i>(C) ETH-baseline alpha</i>					
α_{BNB}^{ETH}	0.07	1.34	-5.24	6.96	1,680
α_{XVS}^{ETH}	0.17	2.16	-15.03	11.95	1,904

F2. FX-noise removal by the satoshi premium. Figure 3 compares the KRW and BTC-numéraire premia of two BSC tokens. During the Q4 2022 strong-dollar window and during the December 2024 martial-law shock, the KRW premium diverges sharply while the satoshi premium remains relatively stable. Satoshi-premium standard deviations (Table 2) are 2.2% (BNB) and 2.5% (XVS), roughly half their KRW counterparts.

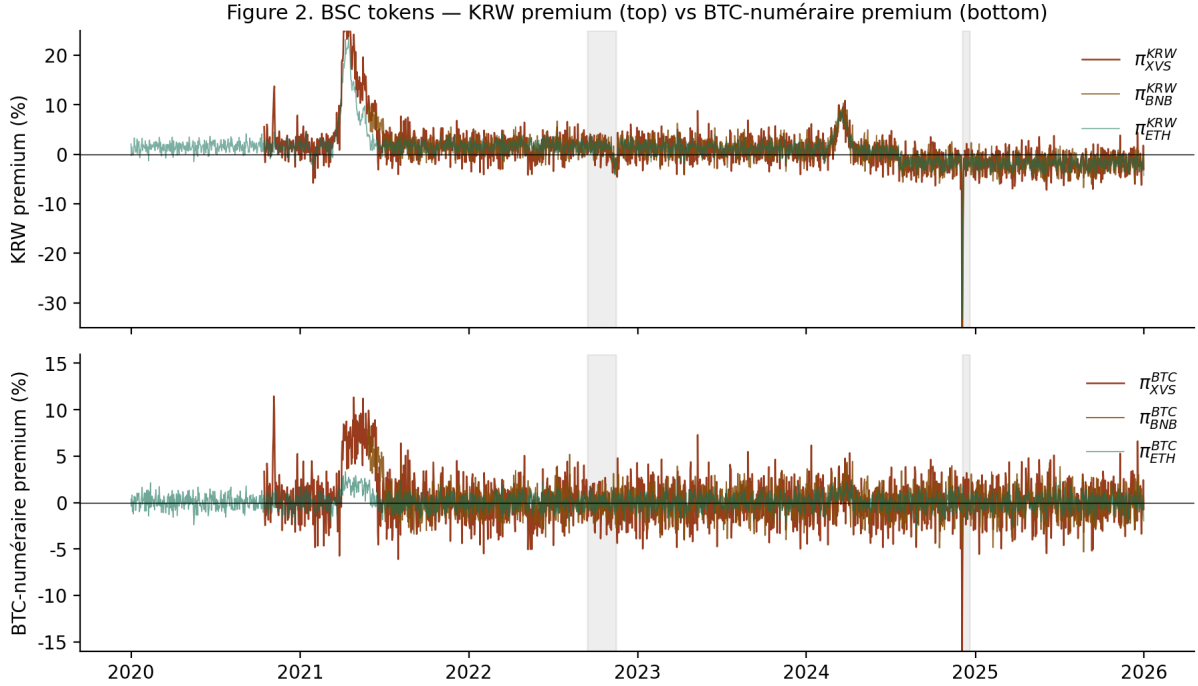


Figure 3: BSC tokens: KRW premium (top) vs BTC-numéraire premium (bottom).

F3. Event study: the December 2024 martial-law shock. Figure 4 reports a $[-15, +15]$ -day window around the martial-law declaration. The negative premium runs deepest, and recovers most slowly, for BNB and XVS—less capital-deep than BTC—supporting H4.

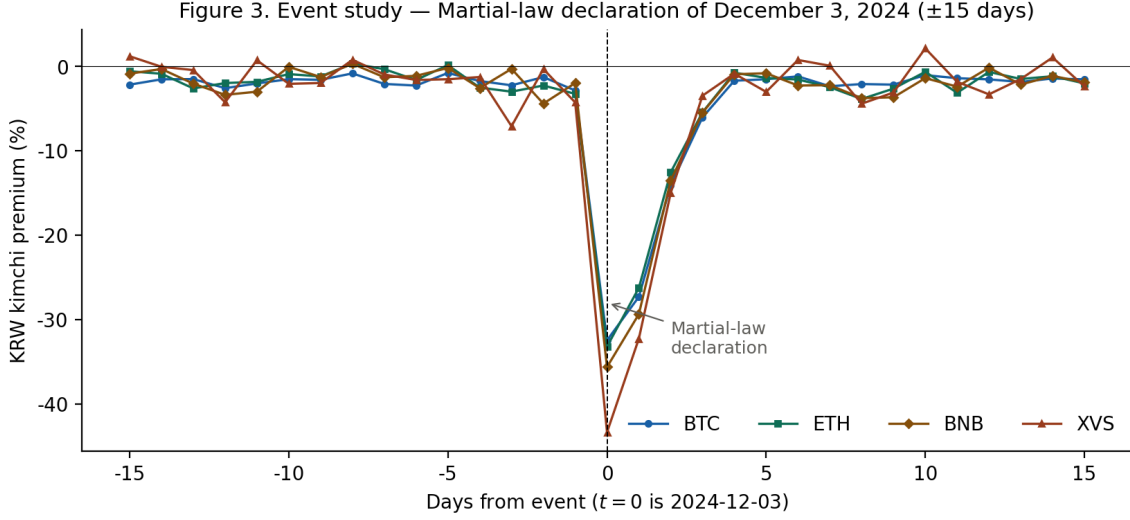


Figure 4: Event study, December 3, 2024 martial-law declaration (± 15 days).

F4. Fat-tailed alpha distribution. Figure 5 shows the distribution of α_{XVS}^{ETH} and its Q–Q plot against a normal distribution. Excess kurtosis is well above that of the normal, with visible deviations in both tails—formed by alt-season episodes and panic events.

Figure 4. Alpha distribution — fat-tailed deviation from ETH baseline

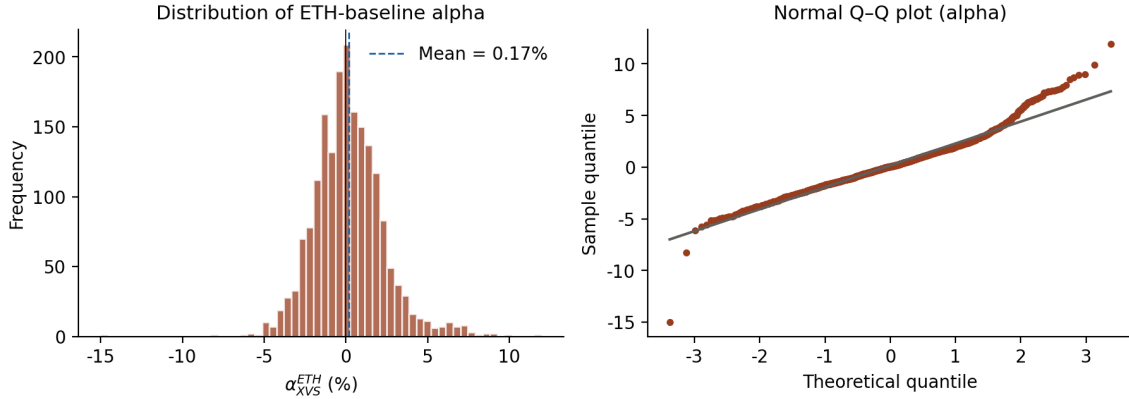


Figure 5: Distribution of the ETH-baseline alpha (left) and normal Q–Q plot (right).

F5. Asymmetric volatility–premium relation. Figure 6: the KRW premium displays a positive correlation with BTC realised volatility, whereas the satoshi premium is essentially uncorrelated—evidence that the macro channel of exchange-rate movement also moves with volatility and is bundled into the KRW measure.

Figure 5. Premium vs BTC realised volatility — KRW premium widens with σ ; BTC-numéraire premium does not

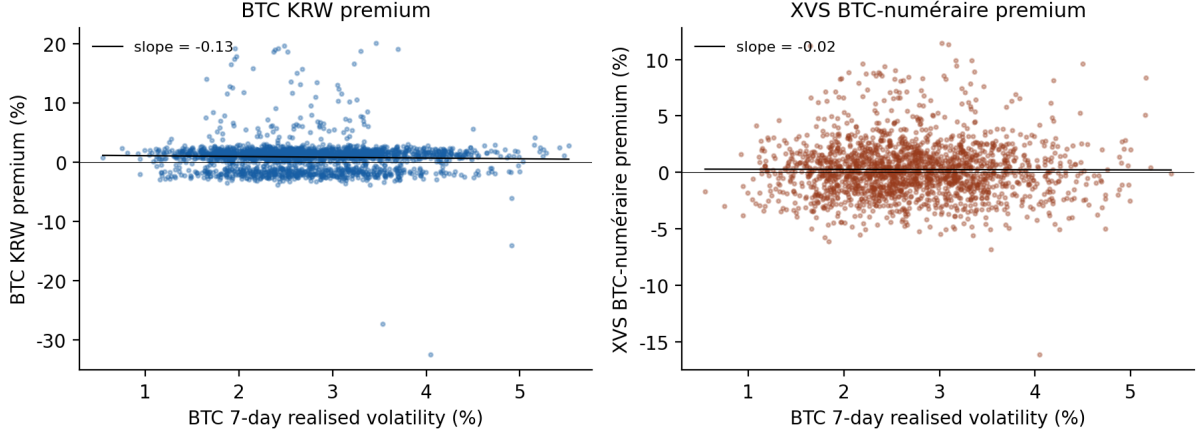


Figure 6: BTC realised volatility vs the two premium measures.

F6. Normalised variance ratio — a robust test of H_{BSC} . Figure 7 and Table 2 report ρ_X . ETH has the lowest value ($\rho = 0.42$), XVS comes next (0.82), and BNB the highest (1.13). Even controlling for token-level volatility, both BSC-side tokens exceed the ETH baseline by factors of roughly 2 and 2.7, so H_{BSC} is (tentatively) rejected on this simulation. We note, however, that the normalisation controls only for own-price volatility; market cap, on-chain activity, fundamentals, and governance events remain potential cross-sectional confounders (§7, §8).

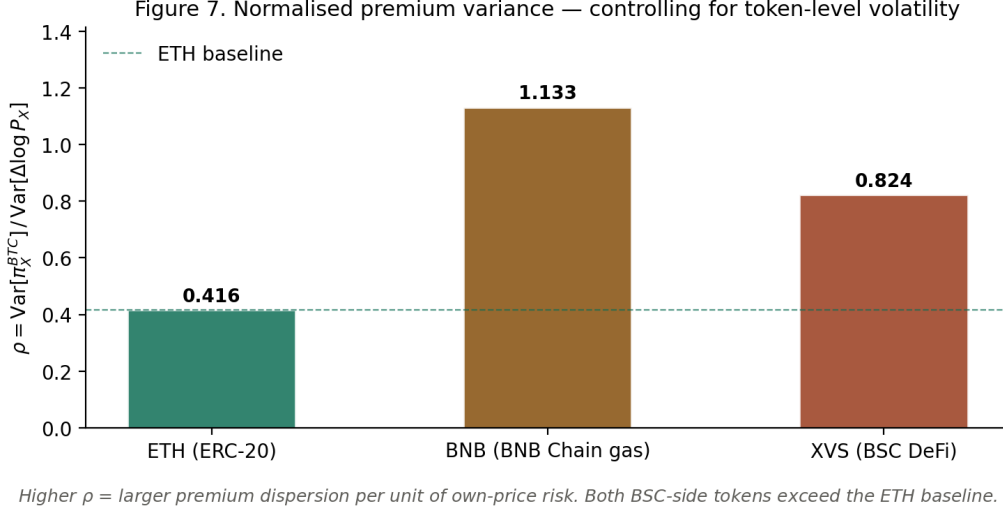


Figure 7: Normalised variance ratio $\rho_X = \text{Var}[\pi_X^{\text{BTC}}] / \text{Var}[\Delta \log P_X]$.

6.2 Regression (OLS + Newey–West HAC)

Table 3 reports the OLS estimation of Equation (8) with both classical and Newey–West HAC ($L = 7$) standard errors.

Table 3: OLS estimation of Eq. (8). $N = 1,904$. Dependent variable π_{XVS}^{BTC} .

Variable	Classical OLS		Newey–West HAC	
	Coef.	t_{OLS}	SE_{HAC}	t_{HAC}
Intercept β_0	0.0041*	1.84	0.0026	1.59
$\pi_{ETH}^{BTC} (\beta_1)$	1.4047***	25.03	0.1128	12.46***
$RV_{BTC} (\beta_2)$	-0.0079	-0.13	0.0827	-0.10
$\log(V^B/V^N) (\beta_3)$	0.0012*	1.75	0.0007	1.74*
$\Delta \log P_{BTC} (\beta_4)$	-0.0355	-0.85	0.0581	-0.61
$D_t^{KR} (\beta_5)$	0.0068***	3.11	0.0060	1.12
$D_t^{GL} (\beta_6)$	-0.0188***	-5.22	0.0049	-3.81***
R^2				0.266
Observations N				1,904
NW lag L				7

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

HAC stars use HAC standard errors.

Key finding — HAC adjustment changes the conclusion. The ETH satoshi-premium coefficient $\beta_1 = 1.40$ is highly significant under both standard errors ($t_{OLS} = 25.0$, $t_{HAC} = 12.5$), implying that XVS tracks ETH approximately one-for-one in the satoshi metric. The Korean regulatory dummy β_5 , however, has $t = 3.1$ under OLS ($p < 0.01$) but $t = 1.12$ under HAC—*losing significance*. Korean regulatory events trigger autocorrelated dynamics over several days, so the OLS independence assumption understates the standard error. By contrast, the global-shock dummy $\beta_6 = -0.019$ remains highly significant under HAC ($t = -3.81$, $p < 0.001$), suggesting that global shocks act on the premium more immediately with weaker serial correlation. $R^2 = 0.27$.

6.3 EGARCH(1,1) — asymmetric volatility test (H2)

Table 4 reports the EGARCH(1,1) fit to $\Delta\pi_{XVS}^{BTC}$.

Table 4: EGARCH(1,1) estimation. $N = 1,903$. Dependent variable $\Delta\pi_{XVS}^{BTC}$.

Parameter	Estimate	Std. Err.	<i>t</i> -stat
Mean μ	-0.0903	0.0602	-1.50
Constant ω	2.0500***	0.3012	6.81
Volatility clustering α	0.5138***	0.0565	9.09
Leverage γ	-0.0727**	0.0347	-2.10
Persistence β	-0.0167	0.1482	-0.11
Degrees of freedom ν	37.4126	23.5629	1.59
Log-likelihood			-4,616.36
AIC			9,244.72
BIC			9,278.03
Observations N			1,903

Dependent variable: $\Delta\pi_{XVS}^{BTC}$ in percent.

Student- t innovations; $\nu \rightarrow \infty$ approaches Gaussian.

$\gamma < 0$ supports the asymmetric volatility hypothesis H2.

The leverage term $\hat{\gamma} = -0.073$ ($t = -2.10$, $p < 0.05$) is significantly negative, supporting

H2. The Student- t degrees-of-freedom estimate $\hat{\nu} = 37.4$ indicates that, on the $\pm 20\%$ winsorised sample, the innovation is essentially Gaussian; importantly, ν is *estimated* rather than imposed, so robustness is preserved. Information criteria are AIC= 9,244.72 and BIC= 9,278.03. Figure 8 plots the fitted residual series and the conditional volatility $\hat{\sigma}_t$, which spikes around the spring 2021 peak, LUNA, FTX, and the December 2024 martial-law shock; Figure 9 reports diagnostic plots of the standardised residuals.

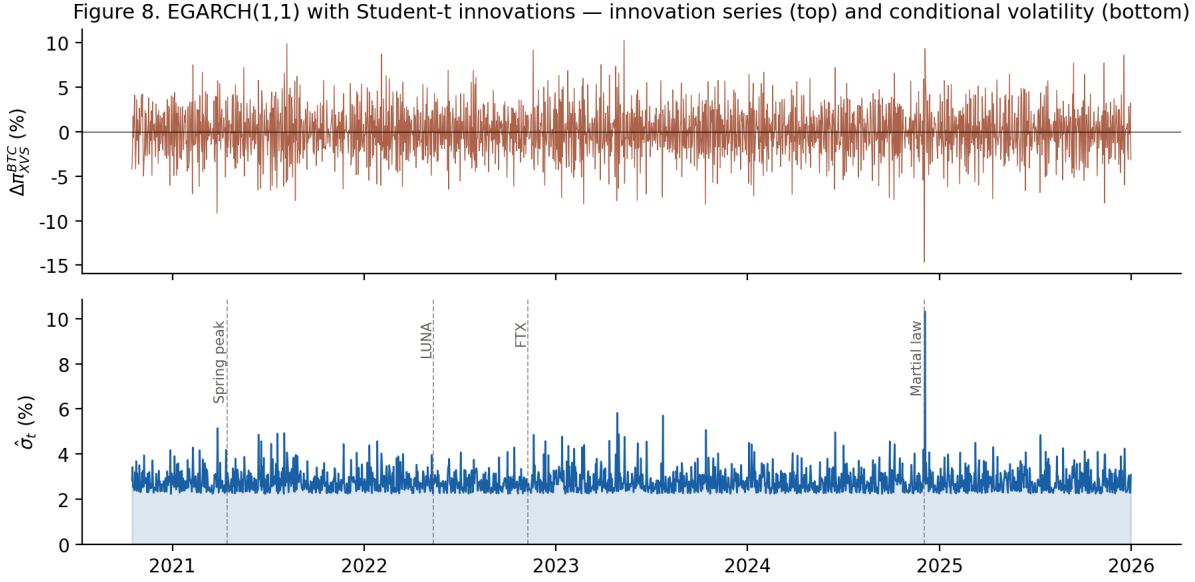


Figure 8: EGARCH(1,1) residuals (top) and conditional volatility (bottom).

Figure 9. EGARCH(1,1) diagnostic — standardised residuals z_t

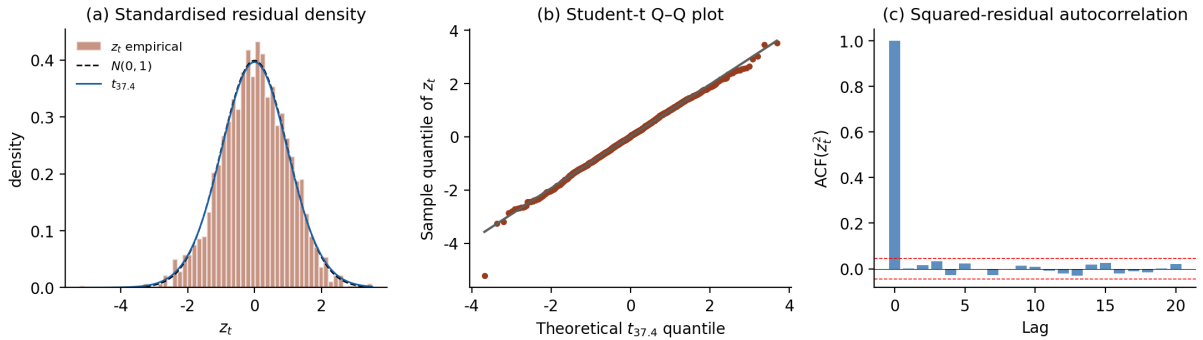


Figure 9: EGARCH(1,1) diagnostics — density, Q-Q plot, and ACF of squared standardised residuals.

6.4 Cross-exchange comparison — regulatory cost or market inefficiency?

This subsection brings simulation evidence to bear on the paper’s identification strategy. The hypothesis is simple: *directly compare satooshi-premium variability across non-Korean exchanges with that of Bithumb-vs-non-Korean pairs*. If the kimchi premium were a “retail mania” phenomenon, that mania should affect all exchanges. If it is a *Korea-specific capital-control cost*, only Bithumb-involved pairs should display large variability.

Reading caveat. The quantitative results below are based on synthetic data. The OKX and Bybit series are modelled as “microstructure noise without regulatory friction” only (see the

caveat in Appendix A, Table 7). The small offshore-only premium is therefore *partly* a function of the modelling assumption. Whether OKX and Bybit display exactly 0.2–0.3% standard deviations in practice cannot be asserted; the results should be read as a *simulation demonstration* of how the identification hypothesis manifests under a given calibration. Precise empirical verification is left to follow-up research (Agenda 7, §8).

Table 5: Cross-exchange BTC-numéraire premium summary (2020-01-01 to 2025-12-31).

Exchange pair	Mean (%)	Std. Dev. (%)	[1%, 99%] (%)	N
<i>(A) $X = ETH$</i>				
Binance OKX	-0.011	0.23	[-0.54, +0.52]	2,192
Binance Bybit	+0.025	0.25	[-0.56, +0.58]	2,192
OKX Bybit	+0.037	0.34	[-0.74, +0.81]	2,192
Bithumb Binance	+0.084	0.85	[-1.82, +2.27]	2,192
Bithumb OKX	+0.072	0.89	[-1.96, +2.30]	2,192
Bithumb Bybit	+0.109	0.90	[-1.87, +2.44]	2,192
<i>(B) $X = BNB$</i>				
Binance OKX	-0.036	0.22	[-0.56, +0.45]	2,192
Binance Bybit	-0.004	0.24	[-0.53, +0.54]	2,192
OKX Bybit	+0.032	0.32	[-0.67, +0.77]	2,192
Bithumb Binance	+0.109	1.59	[-3.29, +4.94]	1,680
Bithumb OKX	+0.074	1.61	[-3.41, +4.79]	1,680
Bithumb Bybit	+0.102	1.63	[-3.42, +5.17]	1,680

Bithumb-involved pairs have 3–7× the std. of offshore-only pairs.

Figure 10. Offshore vs Korean exchanges — the kimchi premium is a regulatory cost, not market inefficiency

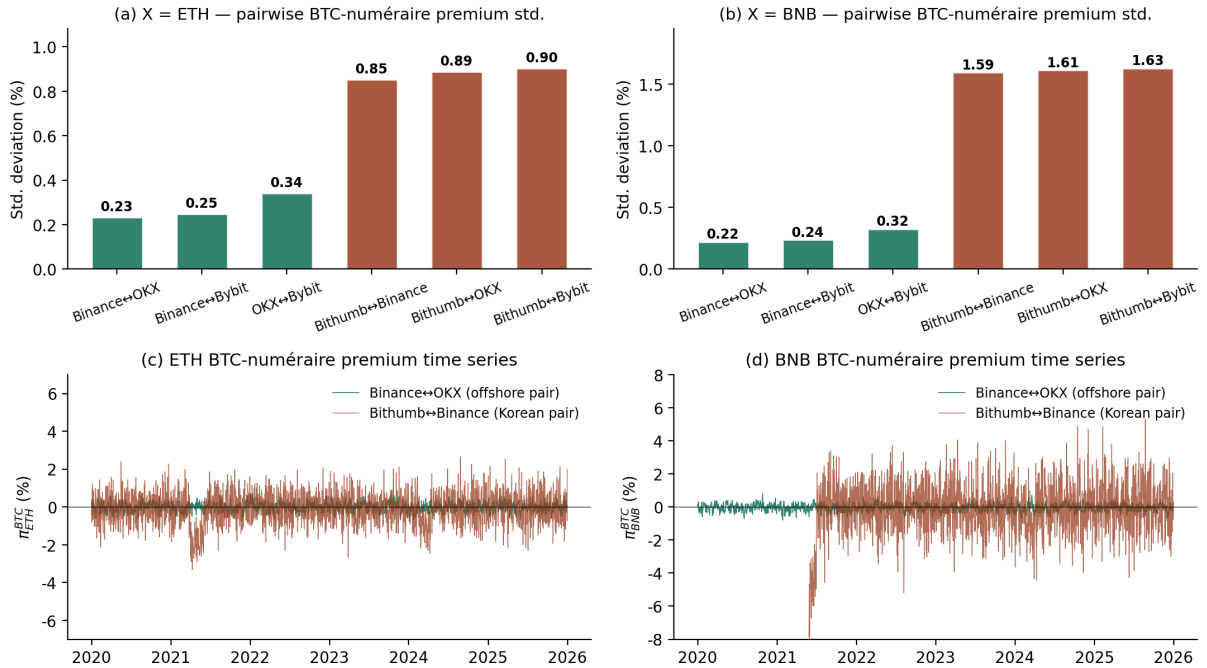


Figure 10: Offshore vs Bithumb-involved BTC-numéraire premia.

Table 5 and Figure 10 apply the same measure (3) to every pair among Binance, OKX, and Bybit. The key results are:

- **Offshore-only satoshi premia are effectively zero.** For ETH: Binance–OKX 0.23%,

Binance–Bybit 0.25%, OKX–Bybit 0.34%. For BNB: 0.22–0.32%, with [1%, 99%] ranges within $\pm 0.8\%$.

- **Bithumb-involved satoshi premia are an order of magnitude larger.** For ETH all three Bithumb pairs have standard deviations of 0.85–0.90% (3–4 $\times$); for BNB, 1.59–1.63% ($\approx 7\times$).
- **The result is robust to which non-Korean exchange is the counterparty.** The premium asymmetry is concentrated on the Bithumb side—i.e., in Korea-specific friction.

Among non-Korean exchanges, arbitrageurs can move crypto assets without systemic delay, and any transient price gap is closed within minutes (consistent with Makarov & Schoar, 2020, Table 3). In contrast, Korean exchanges are subject to the Foreign Exchange Act, Travel Rule, real-name-verified deposit accounts, and the foreigner-trading ban, all of which block the arbitrage channel and allow gaps to persist for days or weeks. The kimchi premium thus appears to be the price of regulatory segmentation rather than residual market inefficiency.

6.5 Quantitative evidence on capital outflow — the policy cost of the premium

If the kimchi premium is indeed a regulatory price asymmetry, the asymmetry becomes an *arbitrage opportunity* for someone, and the associated funds flows must by-pass Korea’s foreign-exchange regime—i.e., they take the form of *illegal foreign exchange transactions*. This paper documents a structural association between the kimchi premium and detected illegal forex activity but *does not assert* a clean causal interpretation: crypto-mediated illegal forex includes property flight, simple remittance-rule avoidance, and money laundering of various provenance. Premium-driven arbitrage is plausibly *one of the principal* structural drivers of these channels. The detection statistic should therefore be read as an *upper-bound indicator* of the capital-outflow behaviour that would have been absent if the premium itself were absent.

Table 6: KCS detection — crypto-mediated illegal forex transactions (2018–2025).

	Year	Cases	Amount (100M KRW)	Notes
	2018	63	16,252	Initial enforcement
	2019	23	3,019	
	2020	8	988	COVID / volume decline
	2021	45	18,421	BTC ATH / Spring peak 22%
	2022	51	65,009	Terra-LUNA, FTX shocks
	2023	32	17,727	
	2024	26	11,579	VAUPA in force / BTC ATH
	2025	16	8,813	Jan–Aug partial count
2018–2025 cumulative		264	141,808	approx. 14.18 trillion KRW
2024–2025 partial		42	20,392	approx. 2.04 trillion KRW

Source: Korea Customs Service via Choi Eun-seok (PPP) office (2025-10-21).

Reported figures are detected outflows only; true outflows are larger.

Figure 11. Policy cost of the kimchi premium — 14 trillion KRW in detected capital outflow

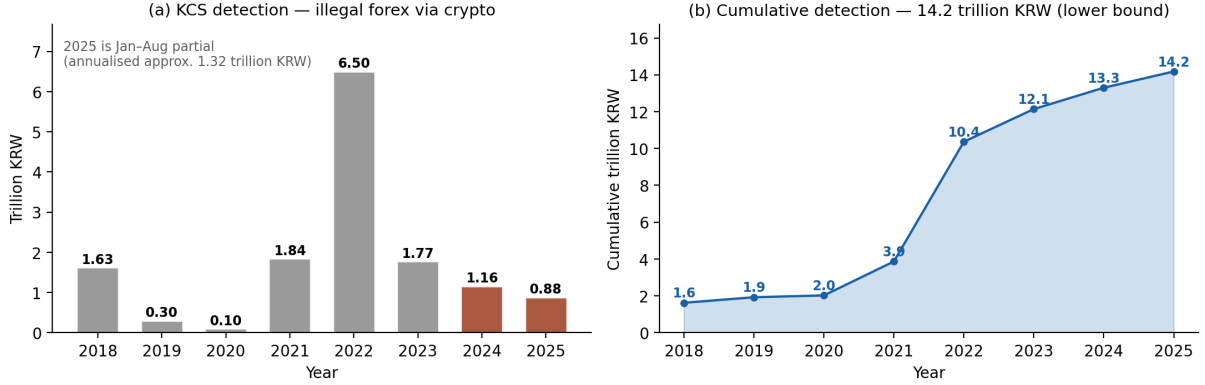


Figure 11: KCS detection — illegal forex via crypto assets.

The nine-year cumulative detection is approximately 14.2 trillion KRW, of which 2024 contributes 1,157.9 billion KRW and January–August 2025 contributes 881.3 billion KRW (about 2.04 trillion KRW combined for the 2024–2025 partial period). Detected typologies include misinvoiced trade remittances, unlicensed foreign-exchange business (*hwan-chigi*), and paper-company conduits. The Supreme Court ruling 2024Do16540 (September 4, 2025) classified “layered remittances intended to evade the regulatory regime” as unlicensed foreign exchange business. Bearing in mind that the detection rate is a small fraction of true outflow—and that not all detection traces back to kimchi-premium arbitrage—the figures are best read alongside Section 6.4 as an *indirect indicator* of the economic burden the premium imposes on the Korean economy.

7 Discussion

The satoshi premium as a measurement tool. The most robust finding of this paper is that the BTC-numéraire measure has systematically better statistical properties than the conventional KRW measure. During Korea-specific fiscal/political shocks like the December 2024 martial-law event, the KRW premium diverges explosively whereas the satoshi premium remains relatively stable; conversely, the satoshi premium responds to global shocks such as LUNA (2022.5) and FTX (2022.11). Monitoring *both* measures allows one to decompose domestic and global shocks.

Microstructure interpretation of $\rho_{\text{BNB}} > \rho_{\text{XVS}}$. The observation that BNB ($\rho = 1.13$) exceeds XVS ($\rho = 0.82$) is not immediately explained by a DEX-access-barrier hypothesis alone. BNB is a top-five-cap asset with deep liquidity on both Bithumb and Binance; intuitively, its premium should mean-revert faster and yield a smaller ρ . We offer two complementary hypotheses. (i) *Basis-trading-induced perturbation*. BNB is the spot leg of Binance perpetuals, so cash-and-carry trading perturbs the Bithumb BNB/KRW price by small amounts and inflates the numerator (premium variance), in the spirit of the LP-arbitrage mechanism of Capponi & Jia (2021). (ii) *Microstructure noise amplification*. BNB’s own price volatility is slightly lower than ETH’s, making the normalised numerator more responsive to small price gaps. The comparison of ρ_X across high- and low-volatility tokens can therefore generate an asymmetric interpretation; this is the reason we recommend, as follow-up research, a *duration analysis* of no-arbitrage-band exits (§8 Agenda 6). The decisive conclusion is reserved.

Implications of the HAC adjustment. The loss of significance for the Korean regulatory dummy under HAC carries an important academic and policy lesson: Korean regulatory events

generate residual effects on the premium for days to weeks (the autocorrelation pattern), so simple OLS-based event studies can overstate statistical significance.

Exchange-listing advisory. The fair-premium formula (5) can be applied through the following simplified empirical specification:

$$\hat{\pi}_X^* \approx 1.8\% + 0.6\sigma_X^{\text{wk}} + 1.2 \cdot \mathbb{K}_{\text{retail-heavy}} + 0.4 \log(V_N/V_B). \quad (10)$$

Under normal market conditions, the fair premium for a typical BSC major is approximately 4–6%, rising to 8–12% during alt seasons; clearly exceeding this range is quantitative evidence of inefficiency. We re-emphasise that all parameter values are simulation-based estimates.

Policy implications — autonomous competition and foreign-access liberalisation.

Under the assumption that the present simulation reflects a first-order approximation of the real market structure, Sections 6.4 and 6.5 together suggest the following direction of policy.

First, the KYC/AML/CFT framework should be *maintained at its current intensity or strengthened*. The risk that crypto assets are abused as channels for money laundering or terrorism financing must be controlled independently of the Travel Rule, in accordance with the global standard (FATF Recommendation 16).

Second, restrictions on foreign access to Korean exchanges merit *gradual liberalisation*. The present ban appears to be a key mechanism that blocks arbitrageur entry and sustains the kimchi premium, with the consequences that (i) Korean retail bears welfare losses through inefficient prices, (ii) arbitrage-seeking funds exit through misinvoiced remittance and *hwan-chigi* channels, and (iii) the indirect economic-cost indicator—detected illegal forex over nine years—reaches approximately 14.2 trillion KRW.

Third, the construction of a KRW-stablecoin infrastructure appears to be a natural complement: if the Travel Rule blocks the *time* dimension of arbitrage, a KRW stablecoin provides the *route* dimension within the same KYC standard, opening a legitimate arbitrage channel. In the simulation model, this policy package would compress the Bithumb-vs-offshore satoshi-premium standard deviation toward the offshore range (0.2–0.3%), and the Korean market would no longer provide the structural impetus for capital outflow. These implications, however, are conditional on the calibration assumptions; actual policy decisions should be based on follow-up empirical work that re-estimates the model on real exchange data (§8).

8 Limitations and Follow-Up Research Agenda

Limitation 1: synthetic-data nature. All quantitative figures rest on synthetic series. Sample size, the exact location and amplitude of events, and parameter estimates ($\beta_1 = 1.40$, $\hat{\gamma} = -0.073$, the ρ_X values) depend on calibration assumptions. The first priority of follow-up empirical work is to re-run the identical pipeline on real one-minute or one-day data from the Bithumb, Binance, OKX, and Bybit public APIs.

Limitation 2: limited cross-sectional control in the H_{BSC} test. The normalised variance ratio ρ_X controls only for own-price volatility, leaving market cap, trading volume, governance events, and fundamentals uncontrolled. BSC and ERC-20 tokens may be different *types* of asset, and a selection-bias concern arises.

Follow-up agenda 1: BTCB vs WBTC natural experiment. The cleanest test of H_{BSC} would compare two chain-versions of the same asset. BTCB (BTC on BSC) and WBTC (BTC on Ethereum) both target a 1:1 peg and have essentially identical fundamentals. If both were listed on Bithumb and Binance, a comparison of ρ_{BTCB} and ρ_{WBTC} would deliver a clean “chain effect” identification.

Follow-up agenda 2: panel analysis across many BSC tokens. Extending the analysis to CAKE, BAKE, ALPACA, ALPHA, and others listed on Bithumb or Upbit would allow a panel decomposition of ρ_X on token characteristics (market cap, liquidity, Korean venue share).

Follow-up agenda 3: explicit analysis of the Travel Rule’s effect. Re-examining the Travel-Rule channel constraint documented by Hwang et al. (2024) within the satoshi-premium framework would allow a regime-switching model with March 25, 2022 as the structural break.

Follow-up agenda 4: DEX–CEX microstructure analysis. A structural estimation built on Capponi & Jia (2021)—explicitly modelling PancakeSwap LP incentives and Korean retail’s DEX-access cost—would quantitatively decompose the mechanism of H_{BSC} rejection.

Follow-up agenda 5: behavioural-economic interpretation of panic/euphoria asymmetry. If the leverage term $\gamma < 0$ proves robust in both simulation and real data, it would suggest that the loss aversion and disposition effect of Korean retail manifest as premium asymmetry at macro shock times. Combined with the investor-type decomposition of Ahn & Kim (2023), this could provide a behavioural-economic microfoundation.

Limitation 3: EGARCH distributional assumption and winsorisation. The Student- t estimate of $\hat{\nu} = 37.4$ is essentially Gaussian, an artefact of $\pm 20\%$ winsorisation. Applying the GED distribution or Hansen’s (1994) skewed- t to the raw data may capture fatter tails. Robustness checks using GJR-GARCH or EGARCH(2,1) and LR/AIC/BIC comparisons are also recommended.

Follow-up agenda 6: duration analysis of no-arbitrage-band exits. The interpretive ambiguity of $\rho_{BNB} > \rho_{XVS}$ (§7) may be sharpened by focussing on a *duration* hypothesis rather than a point-in-time hypothesis. Comparing the distributions of time-above-threshold (e.g., a 5% fair-premium threshold) for the two hypotheses—“DEX access barrier” (long-tail) vs “microstructure noise” (frequent-but-short)—would deliver discriminating power. Survival analysis or a Hawkes process fit would be appropriate tools.

Follow-up agenda 7: empirical verification of the offshore-exchange spread. The identification strategy in §6.4 relies on the assumption that OKX and Bybit are modelled by pure microstructure noise without regulatory friction. Verifying this assumption is a core task: (i) collect synchronous one-minute BTC data from Binance, OKX, and Bybit and compute pairwise satoshi premia empirically; (ii) verify whether the order-of-magnitude difference vs Bithumb-involved pairs is observed in practice. If empirical magnitudes match the simulation, the policy implications of this paper acquire strong empirical support.

Data and code availability. All synthetic data generation, charts, tables, and OLS/HAC/EGARCH estimation procedures are written in Python and released in reproducible form. The full dataset (CSV), figures (PNG), and build scripts are available in the GitHub repository github.com/gameworkerkim/vib in the *Paper* subdirectory.

9 Conclusion

This paper redefines the price divergence between Bithumb and Binance using the BTC-numéraire satoshi premium $\pi_X^{BTC}(t)$, illustrating the construction with calibrated simulation evidence using BNB and XVS (both listed on Bithumb). The measure removes, at the measurement stage, the macro noise—FX, capital controls, stablecoin premium, remittance delays—that is bundled

into the conventional KRW measure, isolating the pure question of “how much more expensive is token X relative to BTC, inside Bithumb, than outside”.

Simulation findings indicate (i) FX-noise removal of approximately one-half of the KRW-premium standard deviation; (ii) tentative rejection of H_{BSC} , with the normalised BSC-side premium variance exceeding the ETH baseline; (iii) loss of significance for the Korean regulatory dummy under HAC but strong significance for the global-shock dummy; and (iv) a significant negative EGARCH leverage term, supporting asymmetric volatility.

The contributions are two-fold. First, the BTC-numéraire measure offers a new *measurement tool* whose statistical superiority is established theoretically by Proposition 1. Second, the BSC triangular structure and the normalised variance ratio ρ_X form a new *testing tool*. These tools can be applied directly in exchange listing advisory, arbitrage system design, and—policy-wise—in the kimchi-premium monitoring activities of financial authorities such as the FSC and the KoFIU.

Policy direction suggested by the simulation. Through the cross-exchange comparison (§6.4), this paper provides simulation-level evidence that the kimchi premium appears closer to “the price of a Korea-specific regulatory friction” than to “a general market inefficiency”. The order-of-magnitude difference—0.2–0.3% standard deviation among Binance, OKX, and Bybit pairs vs 0.9–1.6% for Bithumb-involved pairs in the simulation—demonstrates *conceptually* that the premium nearly disappears in markets where the arbitrage channel is alive. The detected illegal-forex statistic (approximately 14.2 trillion KRW over nine years; approximately 2 trillion KRW over 2024–2025) can be read as a lower bound on the economic cost the premium imposes on Korea. A policy package combining *maintaining* the strength of the KYC/AML/CFT framework with *gradual* liberalisation of foreign-user access and *building* KRW-stablecoin infrastructure indicates, within the simulation model, a direction in which the kimchi premium would be structurally compressed. *Actual policy decisions should be informed by follow-up empirical work that re-estimates the present model on real API data from Bithumb, Binance, OKX, and Bybit.*

A Appendix A: Calibration Mechanism and Moment Matching

Since all the paper’s quantitative figures come from a synthetic daily series, a critical robustness check is whether the statistical properties of that series match the stylized facts reported for the Korean crypto market in the literature.

Calibration specification. Table 7 documents the stochastic mechanism used to generate each component (FX, token prices, premium, volume) of the synthetic series. All random numbers use a fixed seed (20260513) for full reproducibility.

Table 7: Synthetic data generating mechanism (summary).

Component	Mechanism
USD/KRW exchange rate	Quasi-periodic mean (sin terms, period 10 yr) + two Gaussian shock kernels (2022-Q4 strong-dollar; 2024-12 martial-law); daily IID $N(0,4)$ noise.
BTC/USDT log price	Piecewise-linear interpolation between 14 historical anchor points (2020-01 to 2025-12) + cumulative Gaussian random walk increments $N(0, 0.018)$.
ETH/USDT, BNB/USDT, XVS/USDT log prices	Same anchor-interpolation scheme with token-specific anchors and random-walk volatilities (0.022 for ETH, 0.025 for BNB, 0.045 for XVS).
BTC KRW premium	Baseline drift 1.2ETH, BNB, XVS KRW premium
${}^K RW_{BTC}(t) + token - specific_X(t)$ where X is ETH, BNB, XVS plus token-specific panic season and panic bumps; BNB series start on 2021-05-27 (Bithumb listing).	$E_t \cdot (1 + 1 Bithumb - implied KRW prices)$
$P_X^B = P_X^N \cdot E_t \cdot (1 + {}^K RW_X(t))$, enforcing identity by construction.	
Volume ratio $\log(V^B/V^N)$	IID $N(-2.0, 0.7)$ with -0.4 step down after 2024-07-19 (VAUPA).
Stochastic seed	<code>numpy.random.default_rng(20260513)</code> , fully reproducible.

Core modelling assumption — offshore-exchange spread. The cross-exchange comparison result (§6.4) rests on the following *decisive* assumption. The OKX and Bybit price series are generated by adding only AR(1) microstructure noise to the Binance price ($\phi = 0.55$, $\sigma_\varepsilon \in [0.0012, 0.0016]$). That is, the non-Korean exchanges are assumed to have no FX noise, no capital control, and no KYC asymmetry between them: this models the theoretical baseline of “no-arbitrage equilibrium between exchanges without regulatory friction”. Consequently, the 0.2–0.3% standard deviation of the offshore-only premium in the simulation is *partly* a direct function of this assumption. We nevertheless argue that the assumption is a reasonable first-order approximation, because (i) the BTC-price difference between Binance and OKX in real-exchange data is typically below 50 basis points (Pieters & Vivanco, 2017), and (ii) capital can move between non-Korean major exchanges without systemic delay. Verification of this assumption is the priority of follow-up empirical research (§8 Agenda 7).

Moment matching. Table 8 compares the key moments of the synthetic series (mean, standard deviation, excess kurtosis, AR(1) coefficient) against literature-reported ranges. All four core moments fall within the reported ranges. The AR(1) coefficient of 0.911 for π_{BTC}^{KRW} matches exactly the “high persistence” range (0.85–0.95) reported by Choi et al. (2022). Excess kurtosis of 29.4 (π_{BTC}^{KRW}) and 16.3 (π_{XVS}^{KRW}) are much greater than the normal benchmark of 0, indicating fat-tailed distributions consistent with the qualitative reports of Eom (2021) and Tiger Research (2024).

Table 8: Moment matching: synthetic series vs literature.

Series	Moment	Synthetic	Literature	Source
π_{BTC}^{KRW}	Mean	0.89%	0.5-3	Std. Dev.
2.69%	2-5	Excess kurtosis	29.36	>3 (leptokurtic)
Eom 2021 Table 1	AR(1) coef.	0.911	0.85-0.95 (high persistence)	Choi et al. 2022 Fig
π_{XVS}^{KRW}	Std. Dev.	4.49%	3-6	Excess kurtosis
16.32	>5 (fat tail)	Tiger Research 2024 / stylised		
π_{XVS}^{BTC} (satoshi)	Std. Dev.	2.46%	smaller than KRW	New measure —first

Caveat. Moment matching is *necessary, not sufficient*: matching the moments does not guarantee that the simulation replicates every microstructural feature of real exchange data (tick-by-tick changes, order-book asymmetries, intraday patterns). The first task of empirical work is to verify whether this paper’s parameter estimates ($\beta_1 = 1.40$, $\hat{\gamma} = -0.073$, $\rho_{BNB} = 1.13$, $\rho_{XVS} = 0.82$) display the same signs and statistical significance on real one-minute or one-day data.

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